

BIOLOGY PROGRAM ASSESSMENT REPORT

Tracking data

CIM code	MS-BIOL
Program name	Master of Science in Biology
academic year	2024-2025
Program assessment coordinator	Tanya Bekyarova
Assessment plan year	AY24-25
No assessment this year per plan	☐
LO(s) assessed	#4 Communication skills: Demonstrate effective oral and/or written communication skills
Presented at program authority meeting date: <i>Please attach minutes</i>	Biology Department Meeting 10/1/2025

Methodology

Artifacts collected from the two required graduate level Biology courses: Biol 581 Capstone and Biol 533 Graduate Laboratory Techniques, were used to assess LO#4 “**Communication skills:** Demonstrate effective oral and/or written communication skills.” The Biology 581 Capstone Project will be used to assess master's students' oral communication skills, while Biology 533 and the final written report for this class will serve as measures of their written communication skills. The assessment coordinator and the instructor/s of these classes identify student artifacts to be used for assessment. The assessment coordinator collected rubrics, instructor evaluation of student performance, and samples of student work to be used for the assessment.

To assess each learning goal, we will use several performance indicators and three levels of performance for each performance indicator.

Scale for level of performance: 1 = Below expectation, 2 = Meets expectation, 3= Exceeds expectation (Proficient)

Goal	Indicators	1	2	3
4) Communication Skills (oral and written)	- Displays effective written communication skills - Displays effective oral communication skills	Writing is disorganized, with frequent spelling and grammatical errors. Not in scientific format. Disorganized or unable to articulate an argument. Does not grasp intent of the problem.	Well written and organized. Clear & coherent. Engages appropriate audiences	Concise, elegant, engaging. Technical content and graphic design of illustrations well planned / executed. Compelling, persuasive, and accessible to multiple audiences.

Results

Biol 581 Capstone.

Students' oral communication skills in BIOL 581 were evaluated based on their performance in the final in-class oral presentations. To be deemed proficient, students were required to achieve a score of 87% or higher, while those scoring below 70% were classified as not meeting expectations

Rubric for Oral Presentation:

Research Talk Grading Rubric

Criteria	Description	Examples for Scores (1–5)
Research Talk Clarity	Complex ideas are explained clearly and thoroughly.	1: Ideas are confusing and poorly explained. 2: Some ideas are clear, but many are vague or incomplete. 3: Ideas are generally clear with occasional confusion. 4: Most ideas are clearly and thoroughly explained. 5: All ideas are clearly and thoroughly explained with excellent clarity.
Scientific Accuracy	Research is accurate and supported by evidence.	1: Contains major inaccuracies and lacks evidence. 2: Some inaccuracies and limited evidence. 3: Generally accurate with some supporting evidence. 4: Accurate with good supporting evidence. 5: Highly accurate and well-supported by strong evidence.
Engagement	Talk is compelling and encourages questions.	1: Talk is dull and disengaging. 2: Limited engagement with the audience. 3: Moderately engaging with some audience interest. 4: Engaging and encourages some questions.

		5: Highly engaging and stimulates thoughtful questions.
Organization	Well-structured with clear transitions. Slides include: Overview, Background, Rationale & Hypothesis, Methods, Results, Conclusion, Future Prospects.	1: Disorganized with missing components. 2: Some structure but lacks clarity and transitions. 3: Adequate structure with most components present. 4: Well-structured with clear transitions and most components. 5: Excellent structure with all components and seamless transitions.
Verbal Communication	Professional delivery and use of technical language.	1: Unprofessional delivery with poor language use. 2: Somewhat professional but inconsistent language. 3: Generally professional with some technical language. 4: Professional delivery with appropriate technical language. 5: Highly professional and fluent use of technical language.
Preparation and Q/A	Presenter is well-prepared and responds effectively to questions.	1: Unprepared and unable to answer questions. 2: Some preparation but struggles with questions. 3: Moderately prepared and answers most questions. 4: Well-prepared and answers questions effectively. 5: Extremely well-prepared and handles all questions confidently.
Overall Presentation	Overall impression of the talk, including visual aids and delivery.	1: Poor overall impression with weak visuals and delivery. 2: Below average with limited visual support. 3: Average presentation with adequate visuals. 4: Strong presentation with effective visuals and delivery. 5: Outstanding presentation with excellent visuals and delivery.

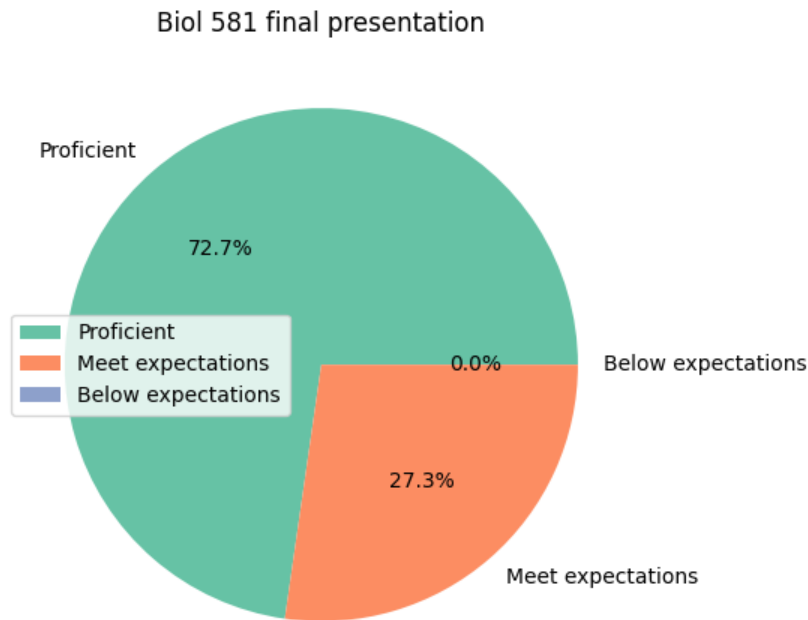


Figure 1: Distribution of evaluations for Biol 581 final presentation. N=11 students

Summary: The data from the final oral presentations in BIOL 581 Capstone indicate that 100% of Biology students successfully demonstrated effective oral communication skills at the Master's level. The evaluation results for the Biol 581 final presentation indicate that the majority of presentations were rated as 'Proficient', with a total of 8 students or 72.7%. A smaller number, 3 evaluations or 27.3%, met expectations, while none were rated as 'Below expectations'. This suggests a strong overall performance by the participants in the class. Samples of student work (Final power point presentation) have been collected.

In BIOL 581, the final oral presentations were evaluated not only by the instructor but also through structured peer review. Peer review scores were incorporated into the students' final grades, providing a multifaceted assessment of their communication skills. The inclusion of peer feedback likely contributed to the high percentage of students achieving proficiency, as it encouraged reflection and refinement of their presentations. For example, several students revised their slides throughout the semester to improve visual clarity and added more precise terminology after receiving peer comments about vague explanations or cluttered graphics. Others adjusted their delivery to better engage the audience, responding to peer suggestions about pacing and tone. The opportunity to revise their oral presentations throughout the semester based on peer input likely contributed to the high percentage of students—72.7%—achieving proficiency on their final presentation. The peer review process fostered a collaborative learning environment and supported the development of higher-quality final submissions through iterative improvement.

Biol 533 Advanced Graduate Laboratory Technique.

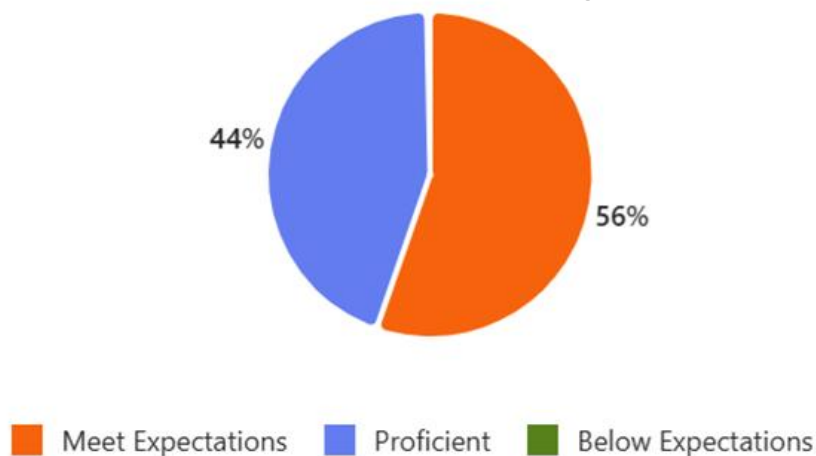
Master's students' written communication skills in BIOL 533 were assessed through their submission of the final written lab report. To be considered proficient, students were required to achieve a score of 88% or higher, while those scoring below 70% were categorized as not meeting expectations.

Rubric for Final lab report:

Final Written Paper Rubric: Communication Skills in Scientific Lab Reporting

Indicators	Does Not Meet	Meets Expectation	Proficient
Scientific Clarity and Accuracy	Scientific concepts are unclear or incorrect; misrepresents experimental work.	Concepts are mostly accurate; minor errors or unclear explanations.	Concepts are clearly and accurately presented; demonstrates strong understanding.
Organization and Structure	Paper lacks logical flow; sections are missing or poorly ordered.	Basic structure present (e.g., intro, methods, results, discussion); some lapses.	Well-organized with clear sections; logical progression of ideas.
Data Presentation and Interpretation	Data is poorly presented or misinterpreted; lacks connection to experiment.	Data is presented with basic interpretation; some connections to experiment.	Data is clearly presented and thoughtfully interpreted; strong analytical links.
Explanation of Experimental Methods	Methods are vague or incomplete; lacks clarity.	Methods are described with some detail; mostly understandable.	Methods are clearly and thoroughly explained; easy to replicate.
Discussion and Analysis	Little to no analysis; fails to connect results to scientific principles.	Basic analysis; some connection to scientific principles and experiment goals.	Deep analysis; clearly connects results to broader scientific context.

Biol 533 Final report



This chart illustrates the distribution of evaluation results across three categories: 'Proficient', 'Meet expectations', and 'Below expectations'. Each segment shows the exact count along with the percentage of total responses.

PERFORMANCE LEVEL	NUMBER OF INDIVIDUALS
Proficient	4
Meet expectations	5
Below expectations	0

Analysis of the final lab papers indicates that 100% of Biology students met the expected standard for written communication at the master's level. Notably, 44% of students exceeded expectations by demonstrating proficiency. Representative samples of student work, in the form of final lab reports, have been collected to illustrate their written communication competencies.

Continuous Improvement, CI, plans

Prior CI plan impact

Refer to the previous assessment report for this LO. Were any CI plans recommended? Were they implemented? Did it work (were improvements seen in this cycle)?

No previous CI plans available

This cycle's CI plan

Describe any CI plans developed from this cycle's data, as well as the strategy developed to implement these plans.

☐ After consideration and presentation to the program authority, no CI plans are recommended.

The data collected from our two graduate classes shows that the students in the Biology department are meeting the expectation of being able to communicate effectively in oral and written form. While all students met expectations, increasing the proportion who exceed expectations will strengthen overall program outcomes and this could be achieved by integrating targeted writing workshops, modules, or mandatory visits to the Communication Laboratory in our MS Program focused on scientific writing conventions, data interpretation, and clarity. Rubrics are already provided and that is why we are focusing on improving students' writing skills by encouraging the students to attend writing workshops or by exploring other campus resources such as the Communication Laboratory that can enhance their writing skills.